

Task 2 — Query Understanding (Decomposition) Report

Database: ClassicModels (PostgreSQL) **Total Questions:** 50 **Format:** Intent · Tables · Columns · Filters · Joins

Part 2A — Simple SELECT Queries (Q1–Q20)

Q1: List all products

Intent: Retrieve all columns and records from the products table **Tables:** products **Columns:** productCode, productName, productLine, productScale, productVendor, productDescription, quantityInStock, buyPrice, MSRP **Filters:** None **Joins:** None

Q2: Get all customers

Intent: Retrieve all columns and records from the customers table **Tables:** customers **Columns:** customerNumber, customerName, contactLastName, contactFirstName, phone, addressLine1, addressLine2, city, state, postalCode, country, salesRepEmployeeNumber, creditLimit **Filters:** None **Joins:** None

Q3: Show all orders

Intent: Retrieve all columns and records from the orders table **Tables:** orders **Columns:** orderNumber, orderDate, requiredDate, shippedDate, status, comments, customerNumber **Filters:** None **Joins:** None

Q4: List all employees

Intent: Retrieve all columns and records from the employees table **Tables:** employees **Columns:** employeeNumber, lastName, firstName, extension, email, officeCode, reportsTo, jobTitle **Filters:** None **Joins:** None

Q5: Get all offices

Intent: Retrieve all columns and records from the offices table **Tables:** offices **Columns:** officeCode, city, phone, addressLine1, addressLine2, state, country, postalCode, territory **Filters:** None **Joins:** None

Q6: Show all product lines

Intent: Retrieve all columns and records from the productlines table **Tables:** productlines **Columns:** productLine, textDescription, htmlDescription, image **Filters:** None **Joins:** None

Q7: List all payments

Intent: Retrieve all columns and records from the payments table **Tables:** payments

Columns: customerNumber, checkNumber, paymentDate, amount **Filters:** None **Joins:** None

Q8: Get product names and prices

Intent: Retrieve the name and buy price of every product **Tables:** products **Columns:** productName, buyPrice **Filters:** None **Joins:** None

Q9: Get customer names and cities

Intent: Retrieve the name and city of every customer **Tables:** customers **Columns:** customerName, city **Filters:** None **Joins:** None

Q10: List employee first and last names

Intent: Retrieve the first name and last name of every employee **Tables:** employees **Columns:** firstName, lastName **Filters:** None **Joins:** None

Q11: Get all order dates

Intent: Retrieve the order number and order date for every order **Tables:** orders **Columns:** orderNumber, orderDate **Filters:** None **Joins:** None

Q12: Show product vendor list

Intent: Retrieve the distinct list of product vendors across all products **Tables:** products **Columns:** productVendor **Filters:** None **Joins:** None

Q13: Get all product codes

Intent: Retrieve the product code of every product **Tables:** products **Columns:** productCode **Filters:** None **Joins:** None

Q14: List all countries from offices

Intent: Retrieve the distinct list of countries where offices are located **Tables:** offices **Columns:** country **Filters:** None **Joins:** None

Q15: Show all order statuses

Intent: Retrieve the distinct list of statuses across all orders **Tables:** orders **Columns:** status

Filters: None **Joins:** None

Q16: Get all payment amounts

Intent: Retrieve the customer number and payment amount for every payment record

Tables: payments **Columns:** customerNumber, amount **Filters:** None **Joins:** None

Q17: List all job titles

Intent: Retrieve the distinct list of job titles held by employees **Tables:** employees **Columns:** jobTitle **Filters:** None **Joins:** None

Q18: Get customer phone numbers

Intent: Retrieve the name and phone number of every customer **Tables:** customers

Columns: customerName, phone **Filters:** None **Joins:** None

Q19: Show product MSRP values

Intent: Retrieve the name and MSRP of every product **Tables:** products **Columns:** productName, MSRP **Filters:** None **Joins:** None

Q20: List order numbers

Intent: Retrieve the order number of every order **Tables:** orders **Columns:** orderNumber **Filters:** None **Joins:** None

Part 2B — JOIN Queries (Q21–Q30)

Q21: Get orders with customer names

Intent: Retrieve each order's number and date along with the name of the customer who placed it **Tables:** orders, customers **Columns:** orders.orderNumber, orders.orderDate, customers.customerName **Filters:** None **Joins:** orders.customerNumber = customers.customerNumber

Q22: Get employees with office city

Intent: Retrieve each employee's first and last name along with the city of their assigned office **Tables:** employees, offices **Columns:** employees.firstName, employees.lastName, offices.city **Filters:** None **Joins:** employees.officeCode = offices.officeCode

Q23: Get payments with customer names

Intent: Retrieve each payment's amount and date along with the name of the customer who made it **Tables:** payments, customers **Columns:** customers.customerName, payments.amount, payments.paymentDate **Filters:** None **Joins:** payments.customerNumber = customers.customerNumber

Q24: Get order details with product names

Intent: Retrieve each order detail record along with the name, quantity ordered, and price of the corresponding product **Tables:** orderdetails, products **Columns:** orderdetails.orderNumber, products.productName, orderdetails.quantityOrdered, orderdetails.priceEach **Filters:** None **Joins:** orderdetails.productCode = products.productCode

Q25: Get products with product line description

Intent: Retrieve each product's name along with the text description of its product line **Tables:** products, productlines **Columns:** products.productName, productlines.textDescription **Filters:** None **Joins:** products.productLine = productlines.productLine

Q26: Get customers with sales rep names

Intent: Retrieve each customer's name along with the first and last name of their assigned sales representative **Tables:** customers, employees **Columns:** customers.customerName, employees.firstName, employees.lastName **Filters:** None **Joins:** customers.salesRepEmployeeNumber = employees.employeeNumber

Q27: Get orders with customer city

Intent: Retrieve each order's number and date along with the city of the customer who placed it **Tables:** orders, customers **Columns:** orders.orderNumber, orders.orderDate, customers.city **Filters:** None **Joins:** orders.customerNumber = customers.customerNumber

Q28: Get employees and their manager

Intent: Retrieve each employee's first name alongside their manager's first name using a self-join on the employees table **Tables:** employees (self-join) **Columns:** employees.firstName, employees.reportsTo, employees.employeeNumber **Filters:** None **Joins:** employees.reportsTo = employees.employeeNumber

Q29: Get orderdetails with product vendor

Intent: Retrieve each order detail record along with the vendor of the corresponding product **Tables:** orderdetails, products **Columns:** orderdetails.orderNumber, products.productVendor, orderdetails.quantityOrdered **Filters:** None **Joins:** orderdetails.productCode = products.productCode

Q30: Get payments with customer country

Intent: Retrieve each payment's amount along with the customer's name and country **Tables:** payments, customers **Columns:** customers.customerName, customers.country, payments.amount **Filters:** None **Joins:** payments.customerNumber = customers.customerNumber

Part 2C — GROUP BY + Aggregation Queries (Q31–Q40)

Q31: Count customers per country

Intent: Count the number of customers grouped by each country using COUNT **Tables:** customers **Columns:** country, customerNumber **Filters:** None **Joins:** None

Q32: Total payments per customer

Intent: Calculate the total payment amount per customer using SUM grouped by customerNumber **Tables:** payments **Columns:** customerNumber, amount **Filters:** None **Joins:** None

Q33: Number of orders per status

Intent: Count the number of orders for each distinct status value using COUNT grouped by status **Tables:** orders **Columns:** status, orderNumber **Filters:** None **Joins:** None

Q34: Products per product line

Intent: Count the number of products in each product line using COUNT grouped by productLine **Tables:** products **Columns:** productLine, productCode **Filters:** None **Joins:** None

Q35: Employees per office

Intent: Count the number of employees in each office using COUNT grouped by officeCode **Tables:** employees **Columns:** officeCode, employeeNumber **Filters:** None **Joins:** None

Q36: Total stock per product vendor

Intent: Sum the quantity in stock for products grouped by each product vendor using SUM **Tables:** products **Columns:** productVendor, quantityInStock **Filters:** None **Joins:** None

Q37: Average buy price per product line

Intent: Calculate the average buy price of products grouped by each product line using AVG
Tables: products **Columns:** productLine, buyPrice **Filters:** None **Joins:** None

Q38: Orders per customer

Intent: Count the number of orders placed by each customer using COUNT grouped by customerNumber **Tables:** orders **Columns:** customerNumber, orderNumber **Filters:** None **Joins:** None

Q39: Max MSRP per product line

Intent: Find the maximum MSRP value among products grouped by each product line using MAX **Tables:** products **Columns:** productLine, MSRP **Filters:** None **Joins:** None

Q40: Min buy price per vendor

Intent: Find the minimum buy price among products grouped by each product vendor using MIN **Tables:** products **Columns:** productVendor, buyPrice **Filters:** None **Joins:** None

Part 2D — Single Aggregate Queries (Q41–Q50)

Q41: Total number of customers

Intent: Count the total number of customer records in the customers table using COUNT
Tables: customers **Columns:** customerNumber **Filters:** None **Joins:** None

Q42: Total number of products

Intent: Count the total number of product records in the products table using COUNT
Tables: products **Columns:** productCode **Filters:** None **Joins:** None

Q43: Total revenue from payments

Intent: Sum the amount across all payment records to get total revenue using SUM **Tables:** payments **Columns:** amount **Filters:** None **Joins:** None

Q44: Average product price

Intent: Calculate the average buy price across all products using AVG **Tables:** products
Columns: buyPrice **Filters:** None **Joins:** None

Q45: Max payment amount

Intent: Find the single largest payment amount across all payment records using MAX

Tables: payments **Columns:** amount **Filters:** None **Joins:** None

Q46: Min payment amount

Intent: Find the single smallest payment amount across all payment records using MIN

Tables: payments **Columns:** amount **Filters:** None **Joins:** None

Q47: Count total orders

Intent: Count the total number of order records in the orders table using COUNT **Tables:** orders **Columns:** orderNumber **Filters:** None **Joins:** None

Q48: Total quantity in stock

Intent: Sum the quantity in stock across all products to get the total inventory using SUM

Tables: products **Columns:** quantityInStock **Filters:** None **Joins:** None

Q49: Average MSRP

Intent: Calculate the average MSRP value across all products using AVG **Tables:** products **Columns:** MSRP **Filters:** None **Joins:** None

Q50: Number of employees

Intent: Count the total number of employee records in the employees table using COUNT

Tables: employees **Columns:** employeeNumber **Filters:** None **Joins:** None

Summary

Category	Questions	Tables Used	Joins Required
Simple SELECT (all columns)	Q1–Q7	1 table each	None
Simple SELECT (specific columns)	Q8–Q20	1 table each	None
JOIN queries	Q21–Q30	2 tables each	Yes (1 join each)
GROUP BY + Aggregation	Q31–Q40	1 table each	None
Single Aggregate (COUNT/SUM/AVG/MAX/MIN)	Q41–Q50	1 table each	None

Total questions decomposed: 50 / 50 **Questions requiring joins:** 10 (Q21–Q30)

Questions using aggregation: 20 (Q31–Q50) **Questions with filters:** 0 (all benchmark questions are unfiltered)